NG CIN SIANG

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STRENGTHS

C | C++ | Python | TypeScript | Docker | Jenkins | Git | Selenium | Playwright | Vision AI | GenAI

SUMMARY

*A senior in the Information Technology Engineering Division of the Department of Civil Engineering, with advanced skills in multiple programming languages and a strong passion for coding. Experienced in **Vision AI** and **e2e test automation**, with hands-on internship experience in building and maintaining **test frameworks**, **automated workflows**, and **CI/CD pipelines**. My research focuses on **Computer Vision** and **Geometric Computation**. Eager to leverage my expertise in **software development** and **automation** to drive impactful solutions.*

WORK EXPERIENCE

Intern, Software Development Engineering in Test — **GoFreight Inc.**
July 2025 - Feb 2026 — Taipei,Taiwan

- Designed an **automated test coverage auditing workflow** integrating GitHub Actions and Jenkins to identify missing test cases across Daily and CI runs, surfacing **50+ uncovered cases** with categorized root causes in Google Sheets. Eliminated **~1 hour** of daily manual triage for the team lead.
- Leveraged AI to semi-automate the migration of **20+ e2e test cases** from Selenium to Playwright, reducing manual effort by **~80%**.
- Maintained a Playwright-based testing framework, resolving pipeline issues across Daily, CI, and e2e runs, and progressively developing new test cases.
- Built and maintained features for an existing e2e system (Python, Selenium, Shell Scripts, Dockerfiles) with **containerized deployment**, ensuring compatibility across GitHub Actions, AWS Spot Instances, and local development environments.

EDUCATION

NATIONAL YANG MING CHIAO TUNG UNIVERSITY(NYCU) — Feb 2024 - Present
MASTER OF SCIENCE MAJOR IN INFORMATION TECHNOLOGY ENGINEERING DIVISION OF THE DEPARTMENT OF CIVIL ENGINEERING — **4.06/4.3 GPA**

NATIONAL YANG MING CHIAO TUNG UNIVERSITY(NYCU) — Sep 2019 - Jan 2024
BACHELOR OF SCIENCE MAJOR IN CIVIL ENGINEERING — **3.63/4.3 GPA**

AWARDS

- **The Outstanding Overseas Chinese Student Scholarship for Graduate Students - 2024–2025 (3 semesters) (top 13** overseas Chinese graduate students chosen by Ministry of Education**)**
- **The Overseas Community Affairs Council Chairman's Award for Outstanding Graduating Overseas Chinese Students - 2023 (top 20** graduate overseas Chinese students chosen by OCAC**)**

THESIS

- **Topology Reconstruction of Distribution Networks Based on Fusion of Deep Learning Recognition from Street and Satellite Imagery**
  - Detected utility poles from **street view** and **satellite imagery** using YOLO11 and YOLO11x models, applying self-constrained Line of Bearing (LOB) intersection for street view localization and GDAL coordinate transformation for satellite-based geolocation.
  - Fused street view and satellite detections using **DBSCAN clustering**, with spatial interpolation to ensure positional continuity of detected poles.
  - Reconstructed **overhead and underground power grid networks** by integrating road network and building footprint data for pole imputation and grid modeling.
  - Designed **road network-constrained topology connections** for power grid modeling, and proposed two **novel topological metrics** alongside APLS to evaluate grid connectivity.

SIDE PROJECT

- **Fall Detection System Using YOLO-Based Skeleton Extraction and a CNN-LSTM FallNet Architecture:**
  A Collaborative Project with Engineers from Hsinchu Science Park